NextAIRE

NextAIRE

Next Generation AI Researchers for Air Quality Excellence

Funded by
the European Union

NextAIRE

AQE – Air Quality and Exposure

Module 2: Air Quality Measurement, Data Quality and QA/QC

Sabina Žero, UNSA, Sarajevo, BiH

Funded by
the European Union

NextAIRE

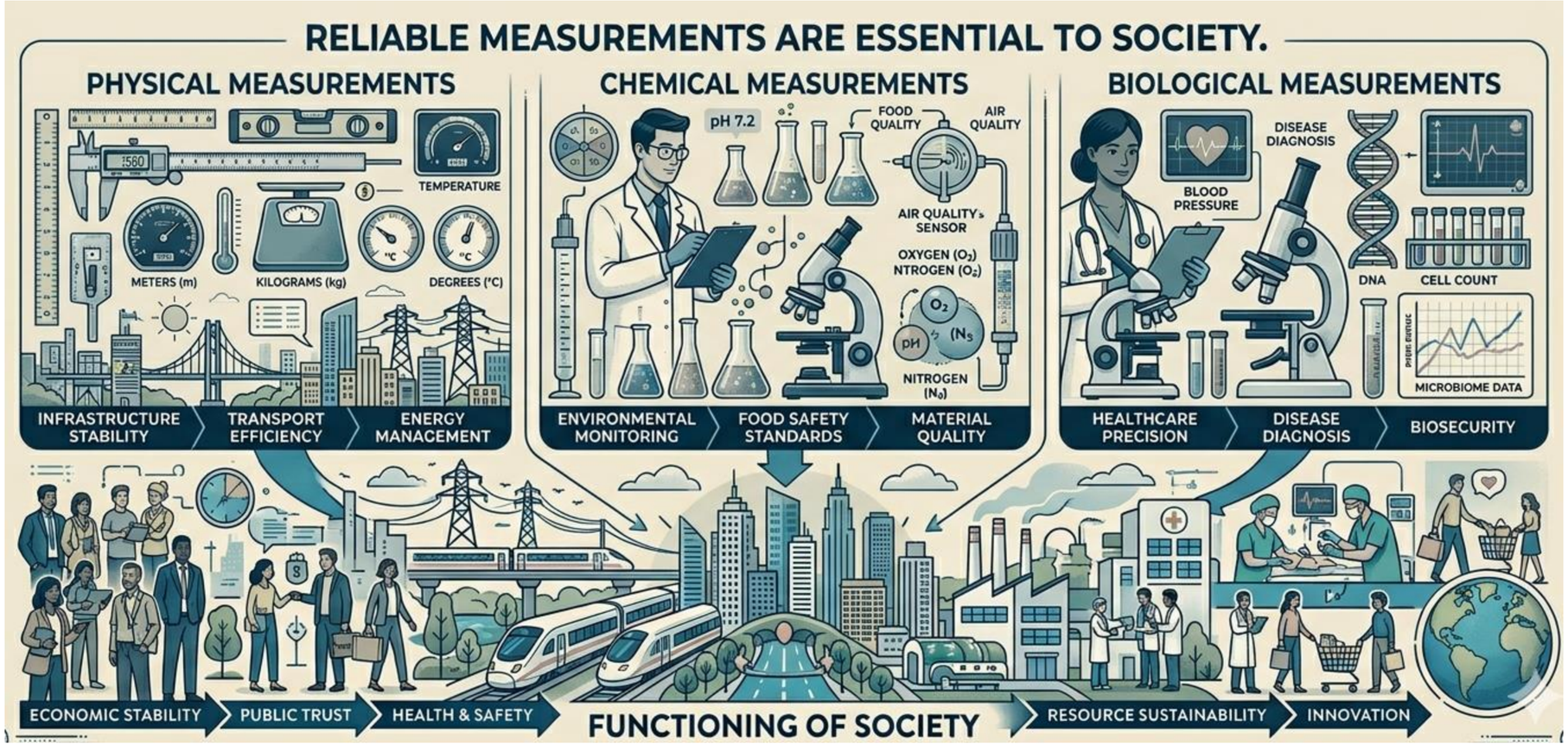
Air Quality Measurement, Data Quality and QA/QC

- The part of the module is focused to provide basic concepts of air quality measurements, QA/QC and metrological traceability

Funded by
the European Union

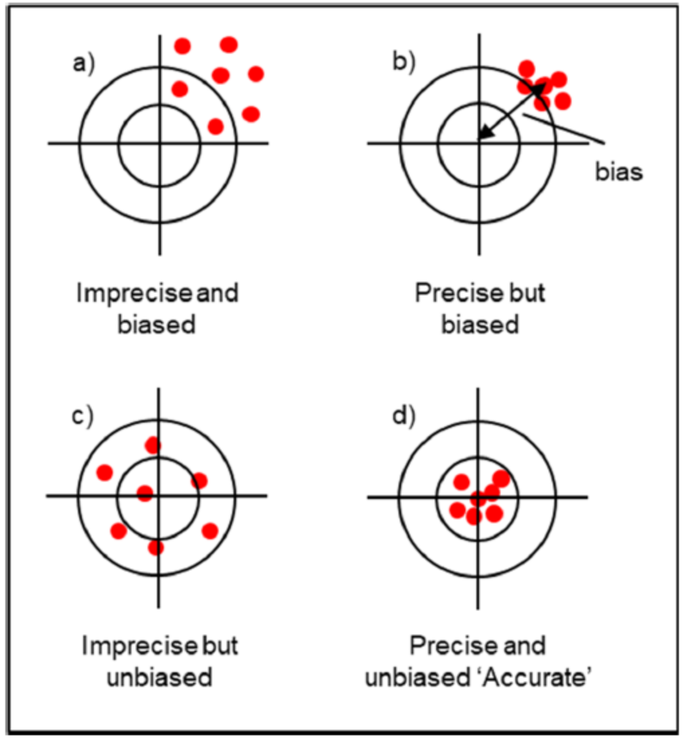
NextAIRE

Measurement



Funded by
the European Union

„We measure, but how?"



Source: V. J. Barwick, 2023, ISBN 978-0-948926-40-2.

Funded by the European Union

Basic terms

Measurement procedure:

- A set of operations, described specifically, used in the performance of particular measurements according to a given method.

Result of a measurement:

- Value attributed to a measurand, obtained by measurement.

Measurand:

- Particular quantity subject to measurement.

Uncertainty of measurement:

- A parameter, associated with the result of a measurement, that characterises
- the dispersion of the values that could reasonably be attributed to the measurand.

Funded by
the European Union

QA/QC

- Quality control (QC) and Quality assurance (QA) are both part of quality management.
- Although they have distinct definitions, QC is often considered to be a subset of QA.

Funded by
the European Union

QA/QC

QUALITY: Degree to which a set of inherent characteristics of an object fulfils requirements (ISO 9000)

QUALITY ASSURANCE (QA): Part of quality management focused on providing confidence that quality requirements will be fulfilled (ISO 9000)

QUALITY CONTROL (QC): Part of quality management focused on fulfilling quality requirements (ISO 9000)

Funded by
the European Union

NextAIRE

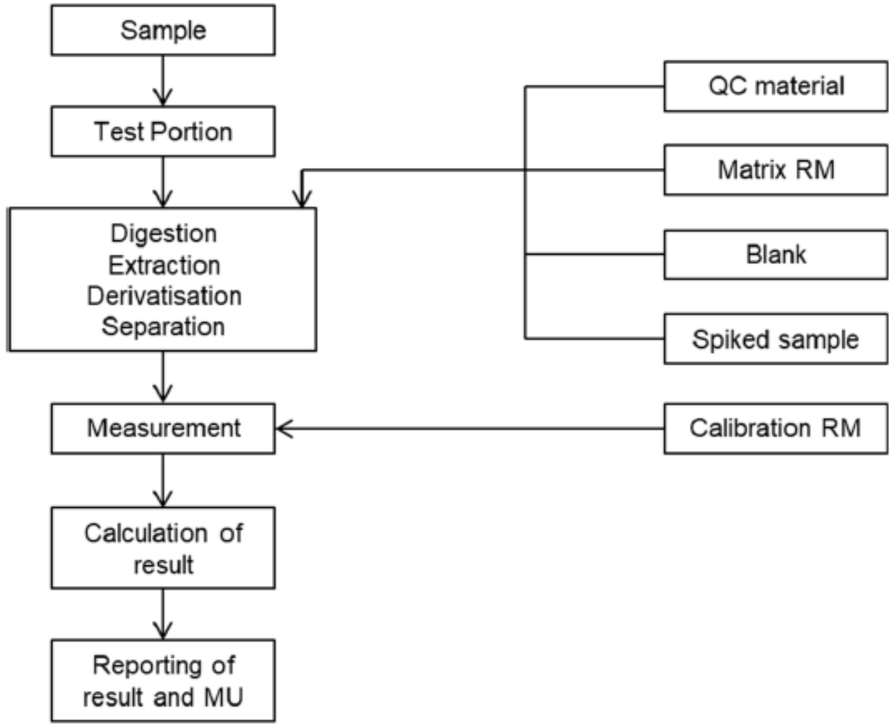
RM/CRM

REFERENCE MATERIAL (RM): Material, sufficiently homogeneous and stable with reference to specified properties, which has been established to be fit for its intended use in measurement or in examination of nominal properties (VIM, 2012)

CERTIFIED REFERENCE MATERIAL (CRM): Reference material, accompanied by documentation issued by an authoritative body and providing one or more specified property values with associated uncertainties and traceabilities, using valid procedures (VIM, 2012)

Funded by
the European Union

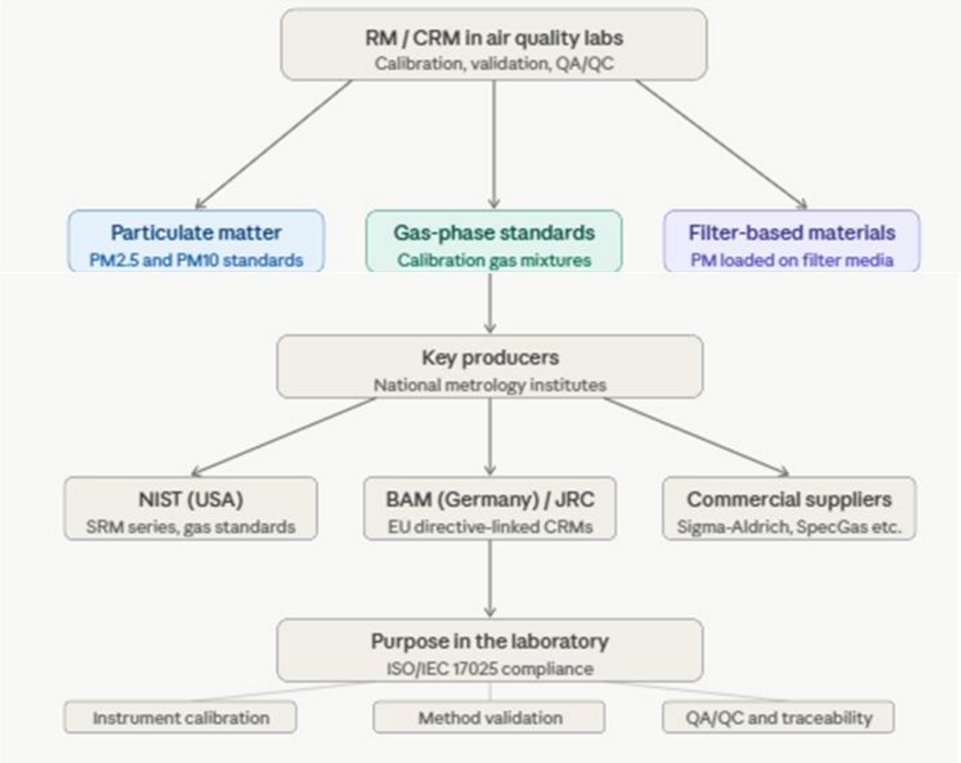
The role of RMs in a typical analytical process



Source: V. Barwick and K. C. Tsimillis, 2026, ISBN 978-0-948926-40-2.

Funded by
the European Union

RM/CRM



Funded by
the European Union

Method validation and verification

Definition	Reference
Verification – confirmation, through the provision of objective evidence, that specified requirements have been fulfilled. **Validation** – confirmation, through the provision of objective evidence, that the requirements for a specific intended use or application have been fulfilled.	ISO 9000:2015[a] [9]
Verification – provision of objective evidence that a given item fulfils specified requirements. **Validation** – verification, where the specified requirements are adequate for an intended use.	ISO/IEC 17025:2017[b] [1]
Verification – confirmation of truthfulness through the provision of objective evidence that specified requirements have been fulfilled. **Validation** – confirmation of plausibility for a specific intended use or application through the provision of objective evidence that specified requirements have been fulfilled.	ISO/IEC 17000:2020 [22]

[a] ISO 9000 notes that the activities carried out for verification are sometimes called a qualification process.

[b] ISO/IEC 17025:2017 refers to the VIM [7] when defining validation and verification.

Source: H. Cantwell (ed.) Eurachem Guide: The Fitness for Purpose of Analytical Methods – A Laboratory Guide to Method Validation and Related Topics, (3rd ed. 2025).

Funded by the European Union

NextAIRE

The performance characteristics in validation

- selectivity
- working range and linearity
- LOD/LOQ
- precision
- trueness
- robustness/ruggedness

Funded by
the European Union

NextAIRE

Metrological Traceability

Property of a measurement result whereby the result can be related to a reference through a documented unbroken chain of calibrations, each contributing to the measurement uncertainty (VIM, 2012)

Funded by
the European Union

Metrological Traceability

Whenever possible, metrological traceability to SI units through suitable measurement standards should be documented, in order to support the comparability of measurement results across space and time.
According to ISO/IEC 17025, this can be achieved through:

- calibration provided by a competent laboratory; or
- certified values of CRMs provided by a competent producer with stated metrological traceability to the SI (RM producers fulfilling the requirements of ISO 17034 are considered to be competent); or
- direct realisation of the SI units ensured by comparison, directly or indirectly, with national or international standards.

Funded by
the European Union

NextAIRE

ILC and PT

In order to monitor the validity of its measurements, it is important for the laboratory to implement a quality assurance (QA) system, which includes the monitoring of its performance by comparison with results of other laboratories, where available and appropriate.

Funded by
the European Union

ILC and PT

Interlaboratory comparison (ILC)

- Organization, performance and evaluation of measurements or tests on the same or similar items by two or more laboratories in accordance with predetermined conditions

Proficiency testing (PT)

- Evaluation of participant performance against pre-established criteria by means of interlaboratory comparisons

Funded by
the European Union

NextAIRE

The Air Quality Standards Regulations' reference methods of measurement for the various pollutants are as specified for the EU so that **Air Quality Measurement, Data Quality and QA/QC** is processed in accordance with the specified methods.

Funded by
the European Union

References

- V. Barwick and K. C. Tsimillis (Eds), Eurachem/CITAC: Guide to Quality in Analytical Chemistry: An Aid to Accreditation (4th ed. 2026). ISBN 978-0-948926-40-2. Available from www.eurachem.org.
- Cantwell H. Eurachem Guide: The Fitness for Purpose of Analytical Methods – A Laboratory Guide to Method Validation and Related Topics, 3rd ed., 2025, www.eurachem.org
- Ramsey M.H., Rostron P.D., Raposo F.C. Eurachem/EUROLAB/CITAC/Nordtest/AMC Guide: "Validation of Measurement Procedures that Include Sampling", Eurachem, 2024, www.eurachem.org
- B. Brookman and I. Mann (eds.) Eurachem Guide: Selection, Use and Interpretation of Proficiency Testing (PT) Schemes (3rd ed. 2021). Available from www.eurachem.org.
- V. J. Barwick (Ed.), Eurachem Guide: Terminology in Analytical Measurement – Introduction to VIM 3 (2nd ed. 2023). ISBN 978-0-948926-40-2. Available from www.eurachem.org.
- Meinrath, G., & Schneider, P. (2007). Quality assurance for chemistry and environmental science: metrology from pH measurement to nuclear waste disposal. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Claude (2026). Images on slides 9,15,20,22,25,26,27,38 [AI-generated image]. Source URL: https://claude.ai
- Gemini (2026). Images on slide 47 [AI-generated image]. Source URL: https://gemini.google.com/

Funded by
the European Union

Are the measurements accurate?

Funded by
the European Union

NextAIRE

Consortium

Ascalia

diadikasia
BRIGHT GROUP MEMBER

smartsense

Megasense

CENTER FOR PHYSICAL SCIENCES AND TECHNOLOGY

INSTITUTE FOR ANTHROPOLOGICAL RESEARCH

theLisbonCouncil
think tank for the 21st century

INAR
Institute for Atmospheric and Earth System Research

imi

msensis

vito

KNOW Center

DHMZ

UNIVERSITAS UNIVERZITET STUDIORUM SARAJEVOENSIS USARAJEVU

DEMOKRITOS

FERIT
FACULTY OF ELECTRICAL ENGINEERING, COMPUTER SCIENCE AND INFORMATION TECHNOLOGY OSIJEK

Funded by
the European Union

Thank You!

Name: Sabina Žero

Organization:
UNSA

Email: sabinazero@pmf.unsa.ba

Website: www.unsa.ba; www.pmf.unsa.ba

NextAIRE

Funded by
the European Union